

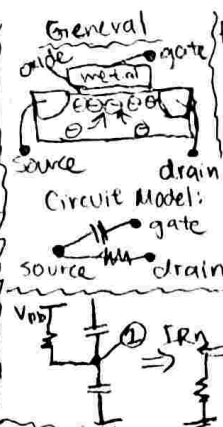
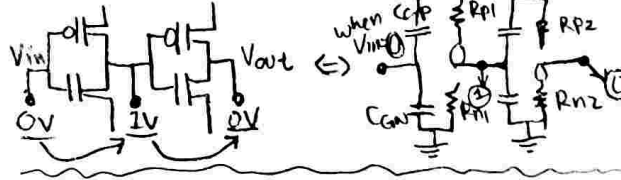
16B Midterm Cheatsheet

Transistors:

NMOS (negative metal oxide semiconductor)

$$G \rightarrow D \Rightarrow \frac{I_D}{V_{GS} - V_{th}} \Rightarrow R_n \Rightarrow V_{GS} \geq V_{th} \text{ where } V_{GS} = (V_{G1} - V_{S1})^*$$

Inverter circuit:



PMOS (positive metal oxide semiconductor)

$$\frac{I_D}{V_{GS} - V_{th}} \Rightarrow \frac{G_D}{C_D} \Rightarrow \text{when } V_{GS} \leq -V_{th} \Rightarrow V_{SG1} \geq |V_{th}|^*$$

Review of R-C circuits:

$$\begin{aligned} (1) I_{C1} &= C_1 \frac{d}{dt}(V(t) - V_{DD}) \\ (2) I_{C2} &= C_2 \frac{d}{dt}(V(t)) \\ (3) I_{RN} &= V(t) / R_n \end{aligned} \Rightarrow \frac{d}{dt} V(t) = -\frac{V(t)}{R_n(C_1 + C_2)}$$

use math to solve differential system
WOO! this for mat looks pretty nice

Review of differential equations:

Homogenous: $\frac{d}{dt} x(t) = \lambda x(t) \Rightarrow x(t) = x(0)e^{\lambda t}$ (capacitor discharging)!

Non homogenous (with a constant): $\frac{d}{dt} x(t) = \lambda x(t) + a$ where $a \neq 0$

hint: (use $\tilde{x}(t) = x(t) + a/\lambda$) $\Rightarrow x(t) = (C_0 - \frac{a}{\lambda})e^{\lambda t} + \frac{a}{\lambda}$ (capacitor charging)!

General overview: $\frac{dV_x}{dt} \begin{cases} -V_x / (R_{on1}P)(C_{G1} + C_{GP}) + V_{DD} / (R_{on1}P)(C_{G1} + C_{GP}) & (V_{in}: 1 \rightarrow 0) \\ -V_x / (R_{on1}N)(C_{GN} + C_{GP}) & (V_{in}: 0 \rightarrow 1) \end{cases}$

Proof of uniqueness:
(1) difference between solutions is 0 across domain where sol is defin
(2) ratio of solutions is always 1. choose an approach.

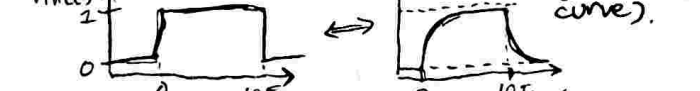
Non homogenous (with a piece wise function)

This derivation involves the use of transforming into discrete time and using $x[i+1] = f(x[i])$ a recursive discrete function and finally taking $\lim_{\Delta t \rightarrow 0}$!!! But you will not worry about this.

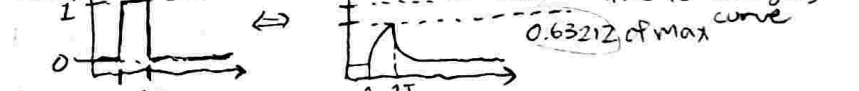
Transient functions and their pitfalls (due to nonzero transistor switching time).

In our model: switch from 0 $\rightarrow$ 1 @ time T then from 1 $\rightarrow$ 0 @ time T + D.

Case 1: $t \in [0, 10T]$



Case 2: $t \in [0, 1T]$



Case 3: input not held long enough to settle.



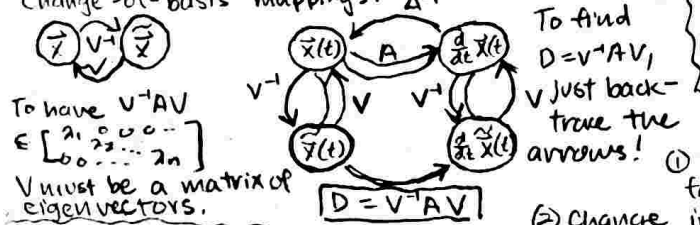
Review of Math 54:

how to solve for $\frac{d}{dt} \vec{x}(t) = A\vec{x}(t) + \vec{b}$
Recall how to find eigenvalues / eigenvectors of a matrix. $A\vec{v} = \lambda\vec{v}$ where $\vec{v}$ is eigenvector and λ is eigenvalue

set $[A - \lambda I]\vec{v} = \vec{0}$ to solve for $\vec{v}$ once eigenvalues found

special cases:
(1) Repeated Eigenvalue: $\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \rightarrow$ eigenspace = $\mathbb{R}^2$! any vector $\vec{v}$ in $\mathbb{R}^2$, $A\vec{v} = \lambda\vec{v}$!
(2) single eigenvalue: $\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \rightarrow$ eigenspace = one dimensional
(3) complex eigenvalues: $\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$ rot. matrix only real @ $\theta = 0^\circ, 180^\circ$ otherwise no real eigenvectors.

How to apply lin alg?



Process of solving systems of DE.

- Find diagonal matrix Λ $\rightarrow$ which can be used to solve for homogenous linear DE in nice basis
- change initial values into other basis and append to front of the homogenous solution.
- change ALL of it back to the original basis.

Inductors and RLC.

$\rightarrow \infty \Omega$
 $I + V^-$
 $V(t) = L \frac{d}{dt} I(t)$
 $E = \frac{1}{2} L I^2$

Stores energy in a magnetic field.
Tips: acts as a short circuit at constant current.

Series equivalence

$$L_{eq} = L_1 + \dots + L_n$$

Recall $L = \frac{N^2 \mu A}{l}$

Parallel equivalence

$$L_{eq} = (L_1^{-1} + L_2^{-1} + \dots)^{-1}$$

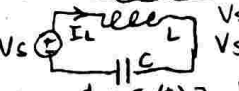
N # of turns

The R-L circuit



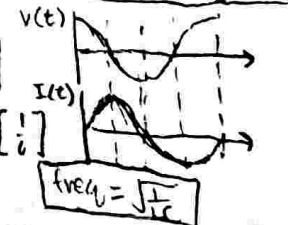
$$\begin{aligned} V_s(t) &= 1V @ t < 0 \\ V_s(t) &= 0V @ t \geq 0 \\ I(t) &= I(0)e^{-Rt/L} \\ V(t) &= RI(0)e^{-t/\tau} \end{aligned}$$

The L-C circuit



$$\begin{aligned} V_s &= 1V @ t < 0 \\ V_s &= 0V @ t > 0 \\ \frac{d}{dt} \begin{bmatrix} I(t) \\ V(t) \end{bmatrix} &= \begin{bmatrix} 0 & -1/L \\ 1/C & 0 \end{bmatrix} \begin{bmatrix} I(t) \\ V(t) \end{bmatrix} \\ \text{Use the stated method} & \lambda_1, \vec{v}_1 = i, [-i] \quad \lambda_2, \vec{v}_2 = -i, [i] \\ \text{use Euler's Identity:} & e^{i\theta} = \cos \theta + j \sin \theta. \end{aligned}$$

$$\begin{bmatrix} I(t) \\ V(t) \end{bmatrix} = \begin{bmatrix} -\sin(\sqrt{\frac{1}{LC}}t) \\ \cos(\sqrt{\frac{1}{LC}}t) \end{bmatrix}$$



A review of complex #s Rectangular form: $z = x + jy$ Polar form $z = |z| e^{j\theta}$
 Euler's identities $\Rightarrow e^{j\theta} = \cos\theta + j\sin\theta$ $\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$ $\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$
 operations in complex are easiest in polar form. where $|z| = \sqrt{x^2 + y^2}$ $\theta = \text{atan}(y/x)$
 // the form $z = |z| e^{j\theta}$ is formed through the Taylor expansion of $e^{j\theta}$

Phasors and impedances

impedances are simply generalizations of resistances.

Sinusoidal inputs for $u(t)$

$v(t) = V_o \cos(\omega t + \phi) = \frac{V_o e^{j\phi}}{2} e^{j\omega t} + \frac{V_o e^{-j\phi}}{2} e^{-j\omega t}$
 // use Euler's to achieve this equality.

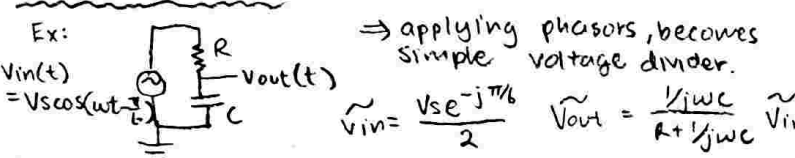
The "Phasor" = $\frac{V_o e^{j\phi}}{2} = \tilde{V}$

Note: The coefficient of exponentials of any of these forms of equations is considered a "phasor"

Inverse-Phasor Transform

$\tilde{W} = \frac{B e^{j\phi}}{2} \rightarrow w(t) = B \cos(\omega t + \phi)$

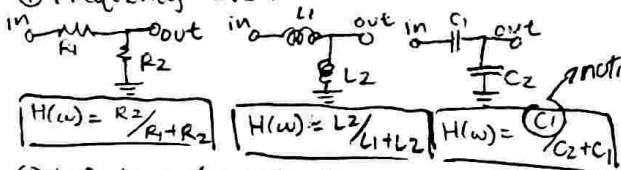
Transfer functions



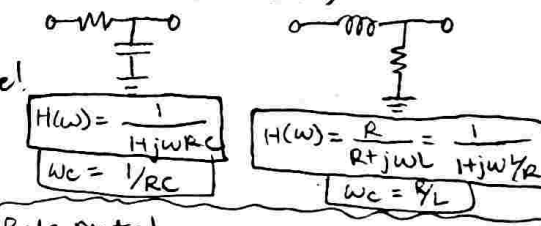
$H(\omega) = \frac{\tilde{V}_{out}}{\tilde{V}_{in}} = \frac{1/j\omega C}{R + 1/j\omega C} \Rightarrow \frac{1}{1 + j\omega RC}$
 // since $H(\omega)$ is a ratio of phasors, they are complex-valued and can be written $H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$
 $|H(\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$ $\angle H(\omega) = -\tan^{-1}(\omega RC)$
 (inverse-phasor transform)

Filters

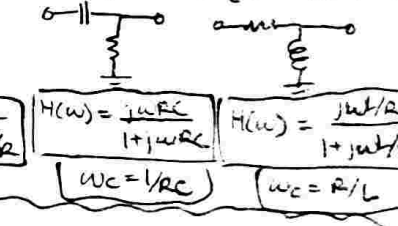
① Frequency-Ind.



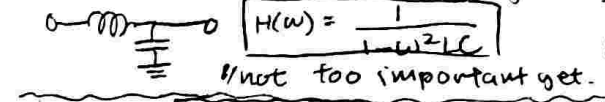
② Low-Pass (RC or LR)



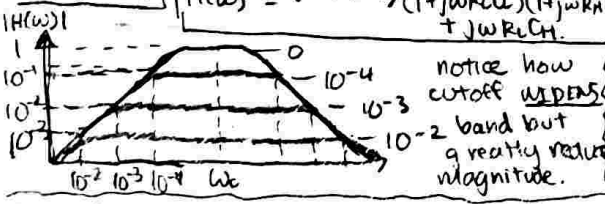
③ High-Pass (CR or RL)



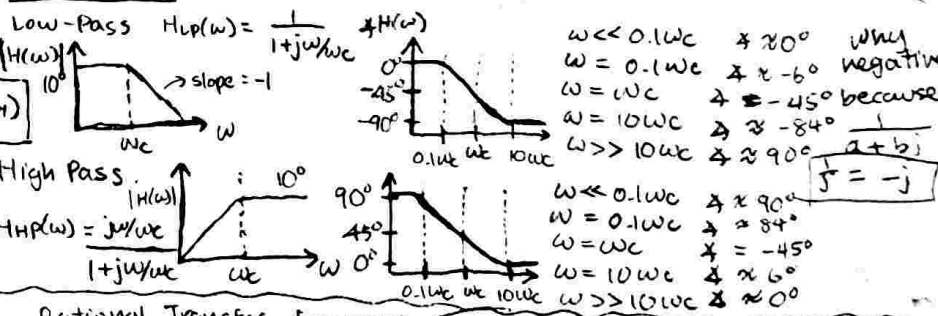
④ L-C tank (constant energy system)



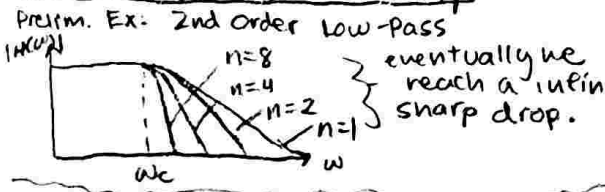
Band Plot



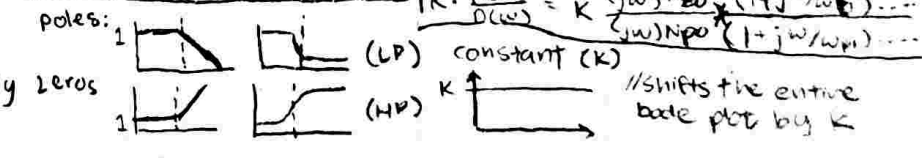
Bode Plots



The effects of higher-ordering



Rational Transfer functions



stability

BIBO stability is a key concept for designing systems. Input (user controlled) bounded results in an output that does not $\rightarrow \infty$ as $t \rightarrow \infty$

Discrete-time vector stability

$\tilde{x}[t+1] = A\tilde{x}[t] + B\tilde{u}[t] + \tilde{w}[t]$
 (user-input)

is stable when $|z_i| < 1$ where z_i is the i^{th} eigenval of A.

How do you prove that even complex eigen vals work? ($|z_i|$ is negative)

Continuous-time stability

$\dot{x}(t) = \lambda x(t) + u(t) + w(t)$
 $x(t) = x(0)e^{\lambda t} + \int_0^t e^{\lambda(t-\theta)} w(\theta) d\theta$

is stable when $\text{Re}\{\lambda\} < 0$

Note 7B system ID.

$\vec{x}[t+1] = A \vec{x}[t] + B \vec{u}[t]$

Let $P = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \dots & \dots & a_{nn} \\ b_{1k} & b_{2k} & \dots & b_{nk} \\ \vdots & \vdots & \ddots & \vdots \\ b_{k1} & \dots & \dots & b_{kn} \end{bmatrix} = \begin{bmatrix} A^T \\ B^T \end{bmatrix}$

$\vec{x}(t)$ and $\vec{x}(t+1)$ are STATE VECTORS
A and B are UNKNOWN
 $\vec{u}(t)$ is our CONTROL VECTOR

$D = \begin{bmatrix} x_1[0] & \dots & x_n[0] & u_1[0] & \dots & u_k[0] \\ x_1[1] & \dots & x_n[1] & u_1[1] & \dots & u_k[1] \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_1[t-1] & \dots & x_n[t-1] & u_1[t-1] & \dots & u_k[t-1] \end{bmatrix}$

$S = \begin{bmatrix} x_1[1] & x_2[1] & \dots & x_n[1] \\ x_1[2] & x_2[2] & \dots & x_n[2] \\ \vdots & \vdots & \ddots & \vdots \\ x_1[t] & x_2[t] & \dots & x_n[t] \end{bmatrix}$

DP $\approx$ S //sanity check!

$P = (D^T D)^{-1} D^T S$ Least-Squares method to solve P

Least-Squares Review:
 $A^T A \vec{x}^* = A^T \vec{b}$ is sol for
 $\vec{x}^* = (A^T A)^{-1} A^T \vec{b}$ $A \vec{x} = \vec{b}$

Note 8 Stability

$\vec{x}[t+1] = A \vec{x}[t] + B \vec{u}[t] + \vec{w}[t]$ //noise.

Example system:

$\vec{x}_d[t+1] = \lambda \vec{x}_d[t] + \vec{w}[t]$

"BIBO stability": for bounded inputs on $\|\vec{u}[t]\| \leq \epsilon$ then $\|\vec{x}_d[t]\| \leq K$ is also bounded.

$\begin{cases} x_d[0] = 0 \\ x_d[1] = w[0] \\ x_d[2] = \lambda w[0] + w[1] \\ x_d[t] = \lambda^{t-1} w[0] + \lambda^{t-2} w[1] + \dots + w[t-1] \end{cases}$ cascading notation for system.

When is this stable?
when $\|\lambda\| \geq 1 \rightarrow$ BIBO unstable
when $\|\lambda\| \leq 1 \Rightarrow \leq \epsilon(1 + \lambda + \lambda^2 + \dots) = \epsilon \frac{1}{1-\lambda} \rightarrow$ BIBO stable

Expand to VECTOR case.

$\vec{x}_d[t+1] = A \vec{x}_d[t] + B \vec{u}[t] + \vec{w}[t]$

$\rightarrow$ decompose $A = V^{-1} \Lambda V$

$\vec{x}_d[t+1] = V^{-1} \Lambda V \vec{x}_d[t] + \vec{w}[t] \Rightarrow \vec{\tilde{x}}_d[t+1] = \Lambda \vec{\tilde{x}}_d[t] + \vec{\tilde{w}}[t]$

$\rightarrow$ let $\vec{u}[0] = 0$ and $\vec{u}[1, 2, \dots] = 0$ (B only takes effect once.)

Question: $\|\vec{w}[t]\| \leq \epsilon$, is $\|\vec{\tilde{w}}\| \leq \epsilon$?

Answer: YES think of it as const * const = const.

$\rightarrow$ Kicker.
all eigenvals of A $\lambda_i < 1$ for system to be BIBO stable

Expand to REALTIME.

$\Rightarrow$ Probably will not be tested on but $\text{Re}\{\lambda\} < 0 \rightarrow$ BIBO stable

Add-on: CLOSED-LOOP CONTROL

Let $\vec{u}_d[i] = K \vec{x}_d[i]$ where K is a matrix WE choose

Take an unstable system

$\vec{x}_d[t+1] = A \vec{x}_d[t] + B \vec{u}_d[t] + \vec{w}[t] \Rightarrow A \vec{x}_d[t] + B K \vec{x}_d[t] + \vec{w}[t]$

where A eigenvals are unstable

$= [A+BK] \vec{x}_d[t] + \vec{w}[t]$ $\rightarrow$ set K so that eigenvals of A+BK are BIBO stable!

$\begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \vec{w}$; let $K = [k_1 \ k_2] \Rightarrow A+BK = \begin{bmatrix} 0 & 1 \\ 3+k_1 & 2+k_2 \end{bmatrix}$ //you have say over state matrix now!

Note 9: Controllability

consider the system $\vec{x}[t+1] = A \vec{x}[t] + B \vec{u}[t]$

assume we want to reach state $\vec{x}[1] = \vec{x}^* \Rightarrow \vec{x}^* = A \vec{x}[0] + B \vec{u}[0] \Rightarrow B \vec{u}[0] = A \vec{x}[0] - \vec{x}^*$

Thus we know to reach $\vec{x}^*$ in ONE timestep, $A \vec{x}[0] - \vec{x}^* \in \text{Range}[B]$.

consider $\vec{x}[2] = A \vec{x}[1] + B \vec{u}[1] = A(A \vec{x}[0] + B \vec{u}[0]) + B \vec{u}[1] = A^2 \vec{x}[0] + AB \vec{u}[0] + B \vec{u}[1]$

$\Rightarrow \vec{x}[2] = A^2 \vec{x}[0] + [B \ AB] \begin{bmatrix} \vec{u}[0] \\ \vec{u}[1] \end{bmatrix}$ //stacked matrix notation.

After n timesteps

$[B \ AB \ \dots \ A^{n-1} B]$ becomes $\begin{bmatrix} B & AB & \dots & A^{n-1} B \end{bmatrix} \begin{bmatrix} \vec{u}[0] \\ \vdots \\ \vec{u}[n-1] \end{bmatrix}$ if $\text{range}[B, AB, \dots, A^{n-1} B] = \mathbb{R}^n$ then system is CONTROLLABLE

Now consider controllable sys

$\vec{x}^* = A^n \vec{x}[0] + [B, AB, \dots, A^{n-1} B] \begin{bmatrix} \vec{u}[0] \\ \vdots \\ \vec{u}[n-1] \end{bmatrix} = A^n \vec{x}[0] + \mathcal{C} \begin{bmatrix} \vec{u}[n-1] \\ \vdots \\ \vec{u}[0] \end{bmatrix}$
 $\Rightarrow \mathcal{C} \begin{bmatrix} \vec{u}[n-1] \\ \vdots \\ \vec{u}[0] \end{bmatrix} = \vec{x}^* - A^n \vec{x}[0]$ //solving using Gaussian elim gives you the sequence of commands necessary to reach $\vec{x}^*$

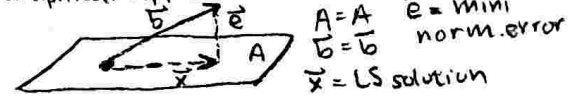
Note 10 Orthonormalization

Review: orthogonality: $\langle \vec{v}, \vec{w} \rangle = 0$
 $\vec{v}^T \vec{w} = \vec{w}^T \vec{v} = 0$

Projections:

$\text{proj}_{\vec{b}} \vec{a} = \frac{\langle \vec{a}, \vec{b} \rangle}{\langle \vec{b}, \vec{b} \rangle} \vec{b}$

Graphical Approach to LS.



Algorithm for orthonormality

original vectors = $\text{span}\{\vec{s}_1, \vec{s}_2, \vec{s}_3, \dots\}$
we want orthonormal vectors = $\{\vec{q}_1, \vec{q}_2, \vec{q}_3, \dots\}$
where $\text{span}\{\vec{s}_1, \vec{s}_2, \vec{s}_3, \dots\} = \text{span}\{\vec{q}_1, \vec{q}_2, \vec{q}_3, \dots\}$
Alg | let $\vec{q}_1 = \vec{s}_1 / \|\vec{s}_1\|$ $\rightarrow$ this is the "starting point"
let $\vec{z}_2 = \vec{s}_2 - (\vec{q}_1^T \vec{s}_2) \vec{q}_1$ and let $\vec{q}_2 = \vec{z}_2 / \|\vec{z}_2\|$
 $\vec{z}_3 = \vec{s}_3 - (\vec{s}_3^T \vec{q}_1) \vec{q}_1 - (\vec{s}_3^T \vec{q}_2) \vec{q}_2$ $\vec{q}_3 = \vec{z}_3 / \|\vec{z}_3\|$
intuition: step 1 creates a vector orthogonal to the other vectors by subtracting components of $\vec{q}_{n-1} \dots \vec{q}_1$ onto $\vec{s}_n$ from $\vec{s}_n$
The "Projection" using inner-product and scalar vector mult.
step 2 is simply normalization.

Note 11 Upper Triangulation $T = \begin{bmatrix} \lambda_1 & * & * & * \\ 0 & \lambda_2 & * & * \\ & 0 & \ddots & * \\ & & & \lambda_n \end{bmatrix}$ $A = UTU^{-1}$ where A is the matrix to triangularize, U is the change of basis to achieve this and T is the triangularized matrix.

$A = U^{-1}AU \Rightarrow \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & * & \\ 0 & \ddots & \lambda_n \end{bmatrix} \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}^{-1} \Rightarrow \begin{matrix} A\vec{v}_1 = \lambda_1 \vec{v}_1 \\ A\vec{v}_2 = * \vec{v}_1 + \lambda_2 \vec{v}_2 \\ \vdots \\ A\vec{v}_n = * \vec{v}_1 + \dots + * \vec{v}_{n-1} + \lambda_n \vec{v}_n \end{matrix}$ evidently, the first vector to the basis U is simple to find.

Guessing the rest of the basis: $R_{n-1} = [\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{n-1}]$ $\star U_n = [\vec{v}_1, R_{n-1}]$ where R_{n-1} is constructed using GS off of $\vec{v}_1$. $\star U_n$ is orthonormal. $\Rightarrow U_n^{-1}AU_n = U_n^T A U_n = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$ Don't care. $\Rightarrow$ can prove $= 0$ using orthonormality. $\Rightarrow$ problematic but overcomeable using induction.

Inductive Reasoning: let A be what we want triangularized. $M_{n-1} = R_{n-1}^T A R_{n-1}$ $[n-1 \times n-1] \rightarrow U_{n-1}^T M_{n-1} U_{n-1}$ is upper-triangular by inductive hypothesis. let $U_n = [\vec{v}_1, R_{n-1} U_{n-1}] \rightarrow$ show $U_n^T U_n = I$ therefore U_n is still orthonormal.

thus $T = U_n^{-1} A U_n = U_n^T A U_n = \begin{bmatrix} \lambda_1 & & \\ 0 & U_{n-1}^T M_{n-1} U_{n-1} \end{bmatrix}$ established as upper-triangular by inductive hypothesis, therefore T is upper-triangular!!!

Schur Decomposition $\rightarrow$ demonstrates that main diagonal still contains eigenvalues! $\Rightarrow \det(A - \lambda I) = \det(T - \lambda I)$ // same characteristic polynomial means same eigenvalues!

SPECTRAL THEOREM!! If A is a SYMMETRIC matrix ($A^T = A$) $\rightarrow$ Then A has real eigenvals and $A = Q^T \Lambda Q$ where Q is an orthogonal eigenvector matrix of A and Λ is a diagonal matrix of eigenvals of A .

Note 12 Minimum Energy Control Recall $\vec{b}_t \vec{u} = \vec{x}^*$ for controllable systems will reach the target $\vec{x}^*$. But how do we minimize the norm of input $\vec{u}$? $\vec{u}$ is entirely orth to $\text{Null}(\vec{b}_t)$ $\vec{a}$ is $\in \text{Null}(\vec{b}_t)$ $Q = D \vec{b}_t$ $D = \vec{b}_t^T$ $Q = \vec{b}_t^T \vec{b}_t$

we can write any solution $\vec{u}_0 = \vec{u} + \vec{a}$ $\rightarrow$ $\vec{u}$ is entirely orth to $\text{Null}(\vec{b}_t)$ $\vec{a}$ is $\in \text{Null}(\vec{b}_t)$ $Q = D \vec{b}_t$ $D = \vec{b}_t^T$ $Q = \vec{b}_t^T \vec{b}_t$

using $Q = \vec{b}_t^T \vec{b}_t \rightarrow$ Prove $\text{Null}(Q) \subseteq \text{Null}(\vec{b}_t)$ and $\text{Null}(\vec{b}_t) \subseteq \text{Null}(Q) \Rightarrow \text{Null}(Q) = \text{Null}(\vec{b}_t)$

Thus, when $Q = \vec{b}_t^T \vec{b}_t$, the orthonormal eigenvectors corresponding to $\lambda = 0$ of Q give $(\vec{v}_{n+1}, \vec{v}_t)$

The first $\vec{v}_1, \dots, \vec{v}_n$ correspond to eigenvals where $\lambda \neq 0$. $\rightarrow$ we can set these arbitrarily for now.

By spectral Thm, $(\vec{v}_1, \dots, \vec{v}_n) \perp (\vec{v}_{n+1}, \vec{v}_t)$

Singular Values $Q \vec{v}_i = \lambda_i \vec{v}_i \Rightarrow \lambda_i = \|\vec{b}_t \vec{v}_i\|^2$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq \lambda_{n+1} = \lambda_{n+2} = \dots = \lambda_t = 0$

set $\lambda_i = \sigma_i^2 \rightarrow \sigma_i = \|\vec{b}_t \vec{v}_i\|$ (singular values of $\vec{b}_t$) $\rightarrow$ simply ordering the eigenvals together

creating an orthonormal basis for $\text{Col}(\vec{b}_t)$, obtain $\vec{w}_1, \dots, \vec{w}_n$ where $\vec{w}_i = \frac{\vec{b}_t \vec{v}_i}{\sigma_i}$ $\Leftrightarrow \vec{b}_t \vec{v}_i = \sigma_i \vec{w}_i$

write out in matrix form:

$\vec{b}_t = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{bmatrix} \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}^{-1}$ append $\vec{v}_{n+1} - \vec{v}_t$ ($\sigma_i = 0$) $\vec{b}_t = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{bmatrix} \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}^{-1}$ (Note: V is orthonormal so $V^{-1} = V^T$)

Clean it up: $\vec{b}_t = W \Sigma V^T$ $\star$ outer product. $\vec{x} \vec{y}^T = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \dots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_1 & x_m y_2 & \dots & x_m y_n \end{bmatrix}$ outer product form of SVD $A = U \Sigma V^T = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{bmatrix} \begin{bmatrix} \vec{v}_1^T \\ \vdots \\ \vec{v}_n^T \end{bmatrix} = \begin{bmatrix} \sigma_1 \vec{u}_1 & & \\ & \ddots & \\ & & \sigma_n \vec{u}_n \end{bmatrix} \begin{bmatrix} \vec{v}_1^T \\ \vdots \\ \vec{v}_n^T \end{bmatrix} = \sigma_1 \vec{u}_1 \vec{v}_1^T + \sigma_2 \vec{u}_2 \vec{v}_2^T + \dots + \sigma_n \vec{u}_n \vec{v}_n^T$

Note 13 SVD $A = [U_r \ U_{m-r}] \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_r^T \\ V_{n-r}^T \end{bmatrix}$

$= \begin{bmatrix} \vec{u}_1 & \dots & \vec{u}_r & \vec{u}_{r+1} & \dots & \vec{u}_m \end{bmatrix} \begin{bmatrix} \sigma_1 & \dots & \sigma_r & 0 & \dots & 0 \\ 0 & \dots & 0 & \sigma_{r+1} & \dots & \sigma_{n-r} \end{bmatrix} \begin{bmatrix} \vec{v}_1^T \\ \vdots \\ \vec{v}_r^T \\ \vec{v}_{r+1}^T \\ \vdots \\ \vec{v}_n^T \end{bmatrix}$ $m \times m$ $m \times n$ $n \times n$

Properties:

- $\text{col}(U_r) = \text{span}\{\vec{u}_1, \dots, \vec{u}_r\} = \text{col}(A)$
- $\text{col}(U_{m-r}) = \text{span}\{\vec{u}_{r+1}, \dots, \vec{u}_m\} \perp \text{col}(A)$
- $\text{col}(V_r) = \text{span}\{\vec{v}_1, \dots, \vec{v}_r\} \perp \text{null}(A)$
- $\text{col}(V_{n-r}) = \text{span}\{\vec{v}_{r+1}, \dots, \vec{v}_n\} = \text{null}(A)$

Space efficiency for SVD. // since most of $\Sigma = 0$, only first r of anything really matters. $A = U_r \Sigma_r V_r^T = \sum_{i=1}^r \sigma_i \vec{u}_i \vec{v}_i^T$

TIP! cols of U form eigenvecs for $A A^T$ rows of V or cols of $V \rightarrow A^T A$ eigenvecs σ_i^2 are eigenvals for one or the other.

Algorithms for finding SVD: ALGORITHMS!

- Find $\lambda_1 \dots \lambda_n$ of $A^T A$ and order them $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} \dots \lambda_n$
- Find corresponding eigenvectors of $A^T A$ $\{\vec{v}_1 \dots \vec{v}_n\}$ // these form the columns of V
- Let $\sigma_i = \sqrt{\lambda_i}$ for $i = 1 \dots \min\{m, n\}$
- Find $\vec{u}_1 \dots \vec{u}_r$ by $\vec{u}_i = \frac{A \vec{v}_i}{\sigma_i}$ for $i = 1 \dots r$
- Find $\vec{u}_{r+1} \dots \vec{u}_m$ using GS orthonormalization.

- Find $\lambda_1 \dots \lambda_n$ of $(A^T A)^T \rightarrow$ different!
- Find Flop. procedure stays same until step 4.
- new Step 4: Find $\vec{v}_1 \dots \vec{v}_r$ by $\vec{v}_i = \frac{A^T \vec{u}_i}{\sigma_i}$ for $i = 1 \dots r$

Note 14: PCA Rooting out non-relevant terms.
 $A = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 2 & 4 & 8 & 10 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ 2/\sqrt{5} & 4/\sqrt{5} \end{bmatrix} \begin{bmatrix} \sqrt{5} & 0 & 0 & 0 \\ 0 & \sqrt{5} & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$
 $\vec{u}_r \rightarrow$ notice, $\vec{u}_r$ is a basis for $\text{col}(A)$!!

PCA in movie recommendations

$\vec{q}_i = \vec{S}_j \cdot \vec{m}_i$ (correlation coefficient). $\vec{m}_i = [a_i \ b_i \ c_i \ d_i]^T$ (feature vector)
 100 people $\vec{a} = [a_{100}]$ 1000 movies $\vec{S}_j = [S_{1000}]$
 consider $\vec{a} \cdot \vec{S}_j^T = [a_{100} \ S_{1000}]$ let $Q = \vec{a} \vec{S}_j^T + \vec{b} \vec{S}_k^T + \dots + \vec{d} \vec{S}_d^T = U \Sigma V^T$
 // outer product
 what are the principle components of Q ?
 along columns $\Rightarrow \vec{u}_1, \vec{u}_2, \dots, \vec{u}_k$
 along rows $\Rightarrow \vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$

Note 15 Linearization

In real-world systems are typically non-linear and continuous.
 Follow the model $\frac{d}{dt} \vec{x} = f(\vec{x}(t), \vec{u}(t)) + \vec{w}(t)$
 drop noise for simplicity.
 $f(x, u) = \frac{\partial f}{\partial x} \big|_{x=x_0} (x-x_0) + \frac{\partial f}{\partial u} \big|_{u=u_0} (u-u_0)$
 Recall Taylor Expansions
 $f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \dots$
 $\approx f(x_0) + \frac{df}{dx} \big|_{x=x_0} (x-x_0)$ // two-term approx.
 more generally $\rightarrow f(\vec{v}) = f(\vec{v}^0) + \sum_{i=1}^n \left(\frac{\partial f}{\partial v_i} \big|_{\vec{v}=\vec{v}^0} \right) (v_i - v_i^0)$
 Vector-valued function linearization

$\vec{f}(\vec{x}, \vec{u}) = \begin{bmatrix} f_1(\vec{x}, \vec{u}) \\ \vdots \\ f_n(\vec{x}, \vec{u}) \end{bmatrix}$
 $f_i(\vec{x}, \vec{u}) = f_i(\vec{x}^*, \vec{u}^*) + \left[\frac{\partial f_i}{\partial x_1} \dots \frac{\partial f_i}{\partial x_n} \right]_{\vec{x}=\vec{x}^*} (\vec{x} - \vec{x}^*) + \left[\frac{\partial f_i}{\partial u_1} \dots \frac{\partial f_i}{\partial u_n} \right]_{\vec{u}=\vec{u}^*} (\vec{u} - \vec{u}^*)$
 for single vars.
 "The Jacobian" $[J_x]$ $\left[\frac{\partial f_1}{\partial x_1} \dots \frac{\partial f_1}{\partial x_n} \right]$ $(\vec{x}^*, \vec{u}^*)$ $[J_u]$ $\left[\frac{\partial f_1}{\partial u_1} \dots \frac{\partial f_1}{\partial u_n} \right]$ $(\vec{x}^*, \vec{u}^*)$
 stuck together

Expansion point choice: choose $(\vec{x}^*, \vec{u}^*) : \vec{f}(\vec{x}^*, \vec{u}^*) = 0$ // "equilibrium point"
 Thus $\frac{d}{dt} \Delta \vec{x}(t) = J_x \Delta \vec{x}(t) + J_u \Delta \vec{u}(t) + \vec{w}(t)$ // linear now!

Note 16 Classification

idea: training set $\vec{x}_1 \dots \vec{x}_n$ indicator (binary) $\ell_1 \dots \ell_n$
 our predictor = $\vec{w}$ this is the correct classification = $\vec{\ell}$ our prediction = $\vec{p}$
 Prediction: $\begin{bmatrix} -\vec{x}_1^T \\ \vdots \\ -\vec{x}_n^T \end{bmatrix} \vec{w} = \vec{\ell}$
 // we can use least squares to solve this! but what is the problem?
 COST FUNCTION // propensity to punish (+3) clearly better and -1 for correct $\ell=1$ is the SAME!
 Prediction $\vec{p}_i = \vec{w}^T \vec{x}_i$ $C(\vec{p}) = (\vec{p} - \vec{\ell})^2$ $C(\vec{p}) = e^{-\vec{\ell} \cdot \vec{p}}$ Goal: find $\vec{w}$.
 Exp. cost function $\arg \min_{\vec{w}} (C(\vec{w}))$ where $C(\vec{w}) = \sum C(\vec{x}_i^T \vec{w})$

Taylor Expansion of derivative of $C(\vec{w})$
 $\frac{dC}{d\vec{w}} \big|_{\vec{w}=\vec{w}^*} + \delta \vec{w} \approx \left[\frac{\partial C}{\partial w_1} \dots \frac{\partial C}{\partial w_n} \right]_{\vec{w}=\vec{w}^*} + (\delta \vec{w})^T \begin{bmatrix} \frac{\partial^2 C}{\partial w_1 \partial w_1} & \dots & \frac{\partial^2 C}{\partial w_1 \partial w_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 C}{\partial w_n \partial w_1} & \dots & \frac{\partial^2 C}{\partial w_n \partial w_n} \end{bmatrix}$
 As a result: (Plugging back into Taylor Expansion)
 $C(\vec{w}^* + \delta \vec{w}) = C(\vec{w}^*) + B |_{\vec{w}=\vec{w}^*} (\delta \vec{w}) + \frac{1}{2} (\delta \vec{w})^T A |_{\vec{w}=\vec{w}^*} (\delta \vec{w})$
 Quadratic! $\leftarrow$ const first-order term 2nd order term
 $A = \begin{bmatrix} \frac{\partial^2 C}{\partial w_1 \partial w_1} & \dots & \frac{\partial^2 C}{\partial w_1 \partial w_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 C}{\partial w_n \partial w_1} & \dots & \frac{\partial^2 C}{\partial w_n \partial w_n} \end{bmatrix}$ Hessian
 $B = \left[\frac{\partial C}{\partial w_1} \dots \frac{\partial C}{\partial w_n} \right]$

Algorithm: (Newton's Method AKA Iterative Method)

- Arbitrarily choose $\vec{w}^* = \vec{w}[0] = \vec{0}$ (can be anywhere)
- Quadraticize around $\vec{w}^*$ $s(\vec{w}) \approx \vec{w}^T A |_{\vec{w}=\vec{w}^*} \vec{w} + B |_{\vec{w}=\vec{w}^*} \vec{w} + C |_{\vec{w}=\vec{w}^*}$
- Find the minimizer of this quadratic and call it $\vec{w}_{\text{new}}^* = \vec{w}[1]$
- Set $\vec{w}^* = \vec{w}[1]$ and repeat step 2-4
- Stop when $\vec{w}[k]$ and $\vec{w}[k+1]$ are very close

Note 17: Discrete Fourier Transforms

$$B_{\text{(DFT matrix)}} = \begin{bmatrix} 1 & e^{j\frac{2\pi}{N}(1-1)} & \dots & e^{j\frac{2\pi}{N}(N-1)(1-1)} \\ 1 & e^{j\frac{2\pi}{N}(1-2)} & \dots & e^{j\frac{2\pi}{N}(N-1)(2-2)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & e^{j\frac{2\pi}{N}(1-(N-1))} & \dots & e^{j\frac{2\pi}{N}(N-1)(N-1)} \end{bmatrix}$$

all orthonormal
all have norm = $\sqrt{N}$

complex # vector properties

$$P_{\vec{u}} \vec{v} = \frac{\langle \vec{v}, \vec{u} \rangle}{\langle \vec{u}, \vec{u} \rangle} \vec{u}$$

$$\|\vec{v}\|^2 = \sum_{all i} v_i v_i^*$$

$$\langle \vec{v}, \vec{u} \rangle = \vec{u}^T \vec{v} = \vec{u}^* \vec{v}$$

$$\langle c\vec{u}, \vec{v} \rangle = c \langle \vec{u}, \vec{v} \rangle$$

where c is some complex scalar

$$\langle \vec{u}, c\vec{v} \rangle = \bar{c} \langle \vec{u}, \vec{v} \rangle$$

$$M^* M = I$$

M is orthonormal complex

U is orthonormal

$$U = \frac{1}{\sqrt{N}} B$$

normalized B

Tip: always try to put $\vec{x}_{\text{test}}$ data in terms of $\vec{u}_1 \dots \vec{u}_{N-1}$!

$$\begin{aligned} \vec{x}_{\text{(test data)}} &= B \vec{\alpha} \\ \vec{\alpha} &= B^{-1} \vec{x}_{\text{test}} = \frac{1}{N} B^* \vec{x}_{\text{test}} \\ B^{-1} &= \frac{1}{N} B^* \quad U^{-1} = U^* \end{aligned}$$

equivalent to simply transposing in the real case

* operator $\rightarrow$ "transpose conjugate"

DFT Review Session:

$$W_N^k = e^{j\frac{2\pi}{N}k}$$

magnitude of all $W_N^k = 1$

$$\text{moving onto } \vec{u}_k = [W_N^{k*0} \ W_N^{k*1} \ W_N^{k*2} \ \dots \ W_N^{k*(N-1)}]^T$$

kth basis vector for DFT

Properties of W_N^k
 $W_N^k = W_N^{-k}$ complex conjugacy

$$W_N^N = 1 = e^{2\pi j}$$

$$W_N^k = W_N^{k+N}$$

periodicity

$$\sum_{k=0}^{N-1} W_N^k = 0$$

sum of vectors (unit) around unit circle

Properties of $\vec{u}_k$

$$\|\vec{u}_k\| = \sqrt{N}$$

$$\vec{u}_k = \vec{u}_{k+N}$$

periodicity

$$\vec{u}_k = \vec{u}_{N-k}$$

complex conjugate

$$\langle \vec{u}_k, \vec{u}_i \rangle = 0 \quad i \neq k$$

orthonormality

moving onto coefficients $\vec{\alpha}$

how much does $\vec{u}_k$ explain test data $\vec{x}_{\text{test}}$?

$$\frac{1}{\sqrt{N}} \begin{bmatrix} -\vec{u}_0^* \\ \vdots \\ -\vec{u}_{N-1}^* \end{bmatrix} \vec{x}_{\text{test}} = \vec{\alpha}$$

Interesting Exam Qs.

Fall 19 #10

a) if $\vec{u} = \alpha \vec{f} + \beta \vec{g}$, c) Prove $\vec{f}$ is real.

$$\vec{u} = \alpha \vec{f} + \beta \vec{g}$$

ring a bell?

$$\vec{f} = \frac{1}{N} B^* \vec{f}$$

(N=4)

$$= \frac{1}{N} \begin{bmatrix} -\vec{u}_0^* \\ \vdots \\ -\vec{u}_{N-1}^* \end{bmatrix} \vec{f}$$

$$\vec{u}_1 = \vec{u}_{N-1} = \vec{u}_3$$

$$f(e^{j\theta}) = F[0] + F[1]e^{jk\theta} + F[2]e^{j2k\theta} + F[3]e^{j3k\theta}$$

interpolation result.

How to choose k_1 and k_2 to make whole thing real?

$$e^{j\theta} + e^{-j\theta} = \text{real}$$

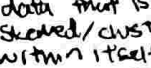
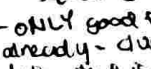
to make whole thing real

Fall 19 #11

$$\text{Square loss: } c^+(p) = (p-1)^2 \quad c^-(p) = (p-1)^2$$

$$\text{Exponential loss: } c^+(p) = e^{-p} \quad c^-(p) = e^p$$

$$\text{logistic loss: } c^+(p) = \ln(1+e^{-p}) \quad c^-(p) = \ln(1+e^p)$$



ONLY good for already clustered data that is not skewed/cluttered within itself.

bad at punishing outliers

Sp 19 #5

let $V_{in}(t) = \cos(\frac{2\pi}{4}t) + \cos(\frac{2\pi}{4}t)$

$$V_{in}(t) = \frac{1}{2} (e^{j\frac{2\pi}{4}t} + e^{-j\frac{2\pi}{4}t}) + \frac{1}{2} (e^{j\frac{2\pi}{4}t} + e^{-j\frac{2\pi}{4}t})$$

$$V_{in}(t) = e^{j\frac{2\pi}{4}t} + e^{-j\frac{2\pi}{4}t}$$

$$V_{in}(t) = \begin{bmatrix} V_{in}(1) \\ V_{in}(2) \\ \vdots \\ V_{in}(k) \end{bmatrix}$$

$$V_{in}(t) = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$V_{in}(t) = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$V_{in}(t) = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Fall 19 #6

we know: to project onto vector

$$P_{\vec{u}} = \frac{\vec{u} \vec{u}^T}{\|\vec{u}\|^2} \text{ or } \frac{\vec{u} \vec{u}^*}{\|\vec{u}\|^2}$$

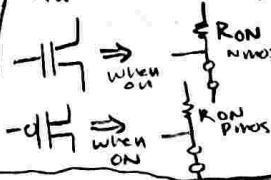
generalize for col Sp(A)

$$P_{\text{col}(A)} = U U^T$$

where U must be a orthonormal basis for col(A). "The equivalent of the unit vector matrix"

Fall 19 #13 Transistors

$$V_{th} = 0.5V \quad V_{tp} = 0.6V$$



always an "ON" resistor to load any current

Sp 19 #9 Minimum Weighted Norm

substitution

want to minimize $\|C\vec{x}\| \leq \|C\vec{z}\|$

$$\text{let } \vec{p} = C\vec{z} \quad \vec{x} = C\vec{z} \quad \|\vec{x}\| \leq \|\vec{p}\|$$

$$\vec{z} = C^{-1}\vec{p} \quad AC^{-1}\vec{p} = \vec{y} \quad AC^{-1}\vec{x} = \vec{y}$$

$$UAC^{-1}\vec{z} = \vec{y} \quad \vec{x} = C\vec{z} = VAC^{-1}\vec{z} = UAC^{-1}\vec{z}$$

$$\vec{x} = C^{-1}VAC^{-1}\vec{z} = UAC^{-1}\vec{z}$$

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